

Note-Continuous and Discrete Adjoints to the Euler Equations for Fluids

Information

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Summary of the content

This article discussed the applications of adjoint equation in compressible flow dynamics which is governed by the Euler equations. The author showed that the continuous and discrete adjoints computed by automatic differentiation agree numerically, and in particular, the adjoint is found to be continuous at the shocks and usually discontinuous at contact discontinuities by both.

Question :

How to find the adjoint equation and the gradient of the objective functional of an incompressible flow governed by Euler equations ?

Consider an optimization problem for an incompressible inviscid Euler flow. The velocity of the fluid is denoted u , the pressure p , the domain occupied by the fluid Ω ; it could be the 2D cross section of a nozzle with an inflow boundary Γ^- , an outflow boundary Γ^+ and walls Γ^0 . One seeks for a nozzle inflow velocity a leading to a velocity u_d in a region of space $D \subset \Omega$:

$$\min_a J(u) = \frac{1}{2} \int_{D \times (0,T)} |u - u_d|^2$$

such that

$$\partial_t u + \nabla \cdot (u \otimes u) + \nabla p = 0$$

$$\nabla \cdot u = 0$$

$$u_{t=0} = 0$$

$$u|_{\Gamma^-} = a$$

$$u \cdot n|_{\Gamma^+} = b$$

$$u \cdot n|_{\Gamma^0} = 0$$

The approach of full Lagrangian functional

Let us introduce a Lagrangian functional such that

$$\mathcal{L}(u, p, a, \beta, \gamma, \lambda) = \frac{1}{2} \int_{D \times (0, T)} (u - u_d)^2 + \int_{\Omega \times (0, T)} \beta \cdot \{\partial_t u + \nabla \cdot (u \otimes u) + \nabla p\} + \int_{\Omega \times (0, T)} \gamma \nabla \cdot u + \int_{\Gamma^- \times (0, T)} \lambda (u - a)$$

We consider the weak variations

$$\tilde{a} = a + \varepsilon \delta a$$

$$\tilde{b} = b + \varepsilon \delta b$$

$$\tilde{p} = p + \varepsilon \delta p$$

$$\tilde{u} = u + \varepsilon \delta u$$

where

$$\delta u \cdot n|_{\Gamma^0} = 0$$

$$\delta u \cdot n|_{\Gamma^+} = \delta b$$

$$\delta u_{t=0} = 0$$

If we consider the functional as a function of ε , meaning

$$\begin{aligned} \mathcal{L}(\varepsilon) &= \mathcal{L}(\tilde{u}, \tilde{p}, \tilde{a}, \gamma, \beta, \lambda) \\ &= \frac{1}{2} \int_{D \times (0, T)} (\tilde{u} - u_d)^2 + \int_{\Omega \times (0, T)} \beta \cdot \{\partial_t \tilde{u} + \nabla \cdot (\tilde{u} \otimes \tilde{u}) + \nabla \tilde{p}\} + \int_{\Omega \times (0, T)} \gamma \nabla \cdot \tilde{u} + \int_{\Gamma^- \times (0, T)} \lambda \cdot (\tilde{u} - \tilde{a}) \\ &= J(\tilde{u}, \tilde{a}) + F(\tilde{u}, \tilde{p}, \tilde{a}, \gamma, \beta, \lambda) \end{aligned}$$

and its first variation is

$$\delta \mathcal{L} = \left. \frac{d\mathcal{L}(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} \varepsilon$$

Substituting the weak variations to the Lagrangian functional, we obtain

$$\left. \frac{dJ}{d\varepsilon} \right|_{\varepsilon=0} = \int_{D \times (0, T)} \{(u + \varepsilon \delta u) \delta u - u_d \delta u\}$$

For the advection (convection) term, we have

$$\begin{aligned}(u + \varepsilon \delta u) \otimes (u + \varepsilon \delta u) &= (u_i + \varepsilon \delta u_i)(u_j + \varepsilon \delta u_j) e_i \otimes e_j \\ &= u \otimes u + \varepsilon u \otimes \delta u + \varepsilon \delta u \otimes u + \varepsilon^2 \delta u \otimes \delta u\end{aligned}$$

where $\{e_i\}$ is a basis in the domain. Thus

$$\begin{aligned}f(\varepsilon) &= \nabla \cdot ((u + \varepsilon \delta u) \otimes (u + \varepsilon \delta u)) \\ &= \nabla \cdot (u \otimes u) + \varepsilon \nabla \cdot (u \otimes \delta u) + \varepsilon \nabla \cdot (\delta u \otimes u) + \varepsilon^2 \nabla \cdot (\delta u \otimes \delta u)\end{aligned}$$

and

$$\left. \frac{df}{d\varepsilon} \right|_{\varepsilon=0} = \nabla \cdot (u \otimes \delta u) + \nabla \cdot (\delta u \otimes u)$$

For the continuity equation (mass conservation), we have

$$\nabla \cdot (u + \delta u) = \nabla \cdot u + \nabla \cdot \delta u = \nabla \cdot \delta u$$

Therefore, the first variation of the Lagrangian functional can be written as

$$\begin{aligned}\delta \mathcal{L} &= \left(\int_{D \times (0, T)} (u - u_d) \cdot \delta u + \int_{\Omega \times (0, T)} \beta \cdot \{ \partial_t \delta u + \nabla \cdot (u \otimes \delta u) + \nabla \cdot (\delta u \otimes u) + \nabla \delta p \} \right. \\ &\quad \left. + \int_{\Omega \times (0, T)} \gamma \nabla \cdot \delta u + \int_{\Gamma^- \times (0, T)} \lambda \cdot (\delta u - \delta a) \right) \varepsilon\end{aligned}$$

Since

$$\gamma \nabla \cdot \delta u = \nabla \cdot (\gamma \delta u) - \delta u \cdot \nabla \gamma$$

thus

$$\begin{aligned}\int_{\Omega \times (0, T)} \gamma \nabla \cdot \delta u &= \int_{\Omega \times (0, T)} \nabla \cdot (\gamma \delta u) - \int_{\Omega \times (0, T)} \delta u \cdot \nabla \gamma \\ &= \int_{\Gamma \times (0, T)} \gamma \delta u \cdot n - \int_{\Omega \times (0, T)} \delta u \cdot \nabla \gamma\end{aligned}$$

where

$$\int_{\Gamma \times (0, T)} \gamma \delta u \cdot n = \int_{\Gamma^- \times (0, T)} \gamma \delta u \cdot n + \int_{\Gamma^0 \times (0, T)} \gamma \delta u \cdot n + \int_{\Gamma^+ \times (0, T)} \gamma \delta u \cdot n$$

We also have

$$\beta \cdot \nabla \delta p = \nabla \cdot (\beta \delta p) - \delta p \nabla \cdot \beta$$

therefore

$$\begin{aligned}\int_{\Omega \times (0, T)} \beta \cdot \nabla \delta p &= \int_{\Omega \times (0, T)} \nabla \cdot (\beta \delta p) - \int_{\Omega \times (0, T)} \delta p \nabla \cdot \beta \\ &= \int_{\Gamma \times (0, T)} \delta p \beta \cdot n - \int_{\Omega \times (0, T)} \delta p \nabla \cdot \beta\end{aligned}$$

and

$$\int_{\Omega \times (0,T)} \gamma \nabla \cdot \delta u = \int_{\Gamma \times (0,T)} \gamma \delta u \cdot n - \int_{\Omega \times (0,T)} \delta u \cdot \nabla \gamma$$

Let us rearrange the terms, it renders us

$$\begin{aligned} \left. \frac{d\mathcal{L}}{d\varepsilon} \right|_{\varepsilon=0} &= \int_{\Omega \times (0,T)} (\mathbf{1}_D(u - u_d) \cdot \delta u + \beta \cdot \{\partial_t \delta u + \nabla \cdot (u \otimes \delta u) + \nabla \cdot (\delta u \otimes u)\} - \delta p \nabla \cdot \beta - \delta u \cdot \nabla \gamma) \\ &\quad + \int_{\Gamma \times (0,T)} \delta p \beta \cdot n + \int_{\Gamma \times (0,T)} \gamma \delta u \cdot n + \int_{\Gamma^- \times (0,T)} \lambda \cdot (\delta u - \delta a) \end{aligned}$$

We are going to develop the expression of advection terms

$$\beta \cdot (\nabla \cdot (u \otimes \delta u)) + \beta \cdot (\nabla \cdot (\delta u \otimes u))$$

For the first term

$$\begin{aligned} \beta \cdot (\nabla \cdot (u \otimes \delta u)) &= \beta_i (u_i \delta u_j), j \\ &= \beta_i (u_{i,j} \delta u_j + u_i \delta u_{j,j}) \\ &= \{(\beta_i u_i)_{,j} - \beta_{i,j} u_i\} \delta u_j + \beta_i u_i \delta u_{j,j} \\ &= \nabla(\beta \cdot u) \cdot \delta u - (\delta u \cdot \nabla) \beta \cdot u + (\beta \cdot u) \nabla \cdot \delta u \end{aligned}$$

Its integral form is

$$\int_{\Omega \times (0,T)} \beta \cdot (\nabla \cdot (u \otimes \delta u)) = \int_{\Omega \times (0,T)} \nabla(\beta \cdot u) \cdot \delta u - (\delta u \cdot \nabla) \beta \cdot u + (\beta \cdot u) \nabla \cdot \delta u$$

Applying the Gauss theorem

$$\begin{aligned} \int_{\Omega \times (0,T)} (\beta \cdot u) \nabla \cdot \delta u &= \int_{\Omega \times (0,T)} \nabla \cdot ((\beta \cdot u) \delta u) - \nabla(\beta \cdot u) \cdot \delta u \\ &= \int_{\Gamma \times (0,T)} (\beta \cdot u) \delta u \cdot n - \int_{\Omega \times (0,T)} \nabla(\beta \cdot u) \cdot \delta u \end{aligned}$$

Thus

$$\int_{\Omega \times (0,T)} \beta \cdot (\nabla \cdot (u \otimes \delta u)) = \int_{\Omega \times (0,T)} -(\delta u \cdot \nabla) \beta \cdot u + \int_{\Gamma \times (0,T)} (\beta \cdot u) \delta u \cdot n$$

where

$$\begin{aligned} (\delta u \cdot \nabla) \beta \cdot u &= \beta_{i,j} u_i \delta u_j \\ &= \beta_{j,i} u_j \delta u_i \\ &= (\nabla \beta)_{i,j}^T u_j \delta u_i \\ &= ((\nabla \beta)^T \cdot u) \cdot \delta u \end{aligned}$$

Therefore

$$\int_{\Omega \times (0,T)} \beta \cdot (\nabla \cdot (u \otimes \delta u)) = \int_{\Omega \times (0,T)} -((\nabla \beta)^T \cdot u) \cdot \delta u + \int_{\Gamma \times (0,T)} (\beta \cdot u) n \cdot \delta u$$

For the second term

$$\begin{aligned} \beta \cdot (\nabla \cdot (\delta u \otimes u)) &= \beta_i (\delta u_{i,j} u_j + \delta u_{i,j} u_j) \\ &= \beta_i \delta u_{i,j} u_j \\ &= (\beta_i \delta u_{i,j}) u_j - \delta u_{i,j} \beta_i u_j \\ &= \nabla(\beta \cdot \delta u) \cdot u - (u \cdot \nabla) \beta \cdot \delta u \end{aligned}$$

and the integral form is

$$\int_{\Omega \times (0,T)} \beta \cdot (\nabla \cdot (\delta u \otimes u)) = \int_{\Omega \times (0,T)} \nabla(\beta \cdot \delta u) \cdot u - (u \cdot \nabla) \beta \cdot \delta u$$

In addition

$$\begin{aligned} \int_{\Omega \times (0,T)} \nabla(\beta \cdot \delta u) \cdot u &= \int_{\Omega \times (0,T)} (\beta_i \delta u_{i,j} u_j)_{,j} - \beta_i \delta u_{i,j} u_{j,j} \\ &= \int_{\Omega \times (0,T)} \nabla \cdot ((\beta \cdot \delta u) u) \\ &= \int_{\Gamma \times (0,T)} (\beta \cdot \delta u) u \cdot n \\ &= \int_{\Gamma \times (0,T)} \beta (u \cdot n) \cdot \delta u \end{aligned}$$

Therefore, after rearranging the terms, we have

$$\begin{aligned} \int_{\Omega \times (0,T)} \beta \cdot (\nabla \cdot (u \otimes \delta u)) + \int_{\Omega \times (0,T)} \beta \cdot (\nabla \cdot (\delta u \otimes u)) &= \int_{\Omega \times (0,T)} -((\nabla \beta)^T \cdot u) \cdot \delta u + \int_{\Gamma \times (0,T)} (\beta \cdot u) n \cdot \delta u \\ &\quad + \int_{\Gamma \times (0,T)} \beta (u \cdot n) \cdot \delta u - \int_{\Omega \times (0,T)} (u \cdot \nabla) \beta \cdot \delta u \end{aligned}$$

As for the time derivative term, we have

$$\beta \cdot \partial_t \delta u = \partial_t (\beta \cdot \delta u) - \delta u \cdot \partial_t \beta$$

and its integral form is

$$\int_{\Omega \times (0,T)} \beta \cdot \partial_t \delta u = \int_{\Omega \times (0,T)} \partial_t (\beta \cdot \delta u) - \delta u \cdot \partial_t \beta$$

where

$$\int_{\Omega \times (0,T)} \partial_t (\beta \cdot \delta u) = \left[\int_{\Omega} \beta \cdot \delta u \right]_{t=0}^{t=T}$$

If we set $\beta(T) = 0$, we will have

$$\int_{\Omega \times (0, T)} \beta \cdot \partial_t \delta u = \int_{\Omega \times (0, T)} -\delta u \cdot \partial_t \beta$$

since $\delta u_{t=0} = 0$.

After regrouping the terms, we obtain the following expression

$$\begin{aligned} \left. \frac{d\mathcal{L}}{d\varepsilon} \right|_{\varepsilon=0} &= \int_{\Omega \times (0, T)} \mathbf{1}_D (u - u_d) \cdot \delta u + \int_{\Omega \times (0, T)} -\partial_t \beta \cdot \delta u + \int_{\Omega \times (0, T)} -((\nabla \beta)^T \cdot u) \delta u \\ &\quad - \int_{\Omega \times (0, T)} (u \cdot \nabla) \beta \cdot \delta u - \int_{\Omega \times (0, T)} \nabla \gamma \cdot \delta u - \int_{\Omega \times (0, T)} \nabla \cdot \beta \delta p \\ &\quad + \int_{\Gamma \times (0, T)} (\beta \cdot u) n \cdot \delta u + \int_{\Gamma \times (0, T)} \beta (u \cdot n) \cdot \delta u + \int_{\Gamma \times (0, T)} \beta \cdot n \delta p \\ &\quad + \int_{\Gamma \times (0, T)} \gamma \delta u \cdot n + \int_{\Gamma^- \times (0, T)} \lambda \cdot (\delta u - \delta a) \end{aligned}$$

Assuming that the Lagrangian functional has extremum, thus we have

$$\delta \mathcal{L} = \left. \frac{d\mathcal{L}}{d\varepsilon} \right|_{\varepsilon=0} \varepsilon = 0$$

which implies the Euler-Lagrange's equations.

The volume integral with respect to δu is

$$I_1 = \int_{\Omega \times (0, T)} (-\partial_t \beta - (\nabla \beta)^T \cdot u - (u \cdot \nabla) \beta - \nabla \gamma + \mathbf{1}_D (u - u_d)) \cdot \delta u$$

The boundary integral with respect to δu is

$$I_2 = \int_{\Gamma \times (0, T)} ((\beta \cdot u) n + \beta (u \cdot n)) \cdot \delta u + \int_{\Gamma^- \times (0, T)} (\lambda + \gamma n) \cdot \delta u$$

The boundary integral with respect to δa is

$$I_3 = \int_{\Gamma^- \times (0, T)} (-\lambda) \cdot \delta a$$

The integral of other terms

$$I_4 = \int_{\Omega \times (0, T)} -\nabla \cdot \beta \delta p + \int_{\Gamma \times (0, T)} \beta \cdot n \delta p + \int_{\Gamma^+ \times (0, T)} \gamma \delta b$$

Therefore, at each instant $t \in (0, T)$, the Euler-Lagrange's equations are

$$\begin{aligned}
-\partial_t \beta - (\nabla \beta)^T \cdot u - (u \cdot \nabla) \beta - \nabla \gamma + \mathbf{1}_D(u - u_d) &= 0 & \text{in } \Omega \\
(\beta \cdot u)n + \beta(u \cdot n) &= 0 & \text{on } \Gamma^+ \cup \Gamma^0 \\
(\beta \cdot a)n + \beta(a \cdot n) + \lambda + \gamma n &= 0 & \text{on } \Gamma^- \\
\nabla \cdot \beta &= 0 & \text{in } \Omega \\
\beta \cdot n &= 0 & \text{on } \Gamma \\
\delta b &= 0 & \text{on } \Gamma^+
\end{aligned}$$

thus

$$\partial_t \beta + (\nabla \beta)^T \cdot u + (u \cdot \nabla) \beta + \nabla \gamma = \mathbf{1}_D(u - u_d) \quad \text{in } \Omega$$

and

$$-\lambda = 0 \quad \text{on } \Gamma^-$$

which is the optimality condition.

On the other hand, the adjoint equations imply the Lagrange multipliers such that

$$\delta \mathcal{L} = \nabla_a \mathcal{L} \cdot \delta a = \nabla_a J \cdot \delta a = \delta J$$

Therefore, the gradient of the objective functional is

$$\nabla_a J = -\lambda = \gamma n + (\beta \cdot a)n + \beta(a \cdot n)$$

Thought

If we set

$$\begin{aligned}
\beta &= u^* \\
\gamma n &= p^* n - 2(u^* \cdot a)n
\end{aligned}$$

then we will recover the expression provided in the paper

$$\delta J = \nabla_a J \cdot \delta a = \int_{\Gamma^- \times (0, T)} ((a \cdot n)u^* + (p^* - a \cdot u^*)n) \cdot \delta u$$

What do I learn/understand ?

- When we need to make the connection between a boundary and a volume integral, the first tool we need to consider is the Gauss Theorem (for divergence)/ Stokes Theorem (for rotational) .
- If the boundary condition is a control variable, then we must treat it as a constraint and

impose a Lagrangian multiplier. However we cannot write something like

$$\delta u|_{\Gamma \times (0,T)} = \delta a$$

if we are not going to use sensitivity equations. In our full Lagrangian approach, the sensitivity equations are not used because it was difficult to substitute them into the objective functional.

Reference/Related Articles

- Mark Kot, *A First Course in the Calculus of Variations*, AMS
- https://www.ece.iastate.edu/~namrata/EE520/Calculus_of_Variations.pdf